CIUK 2026, UCL challenge

Which patients' E. coli infections resist co-amoxiclav?

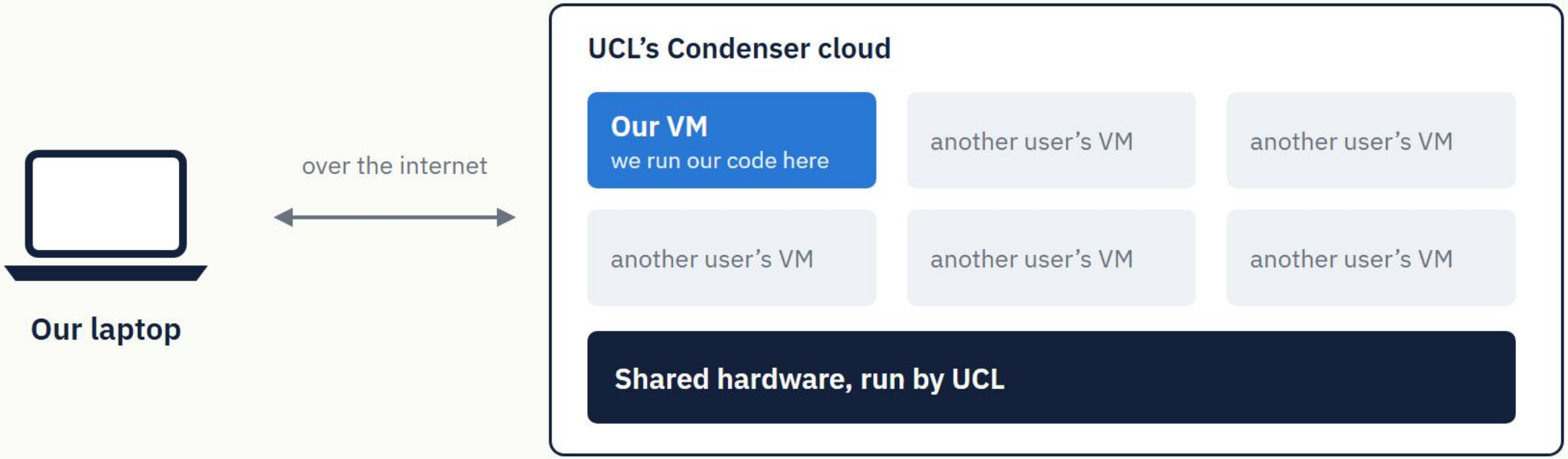
Tree models, run on UCL's Condenser cloud

UCLuster Too

Ada Ge · Bobby Ta · Eric Huang · James Loke

Q1. What is cloud compute?

Cloud computing means renting computers over the internet instead of owning them



Condenser is UCL's private cloud. UCL runs the hardware and we run our code, a split called infrastructure as a service.

Q2. What are key data security considerations for cloud computing?

Who can reach the machine, how data travels, and where the keys are kept



No passwords or keys in our public GitHub repo

The only way in. Our team key is stored here, and only we can read it

In UCL's own facilities, kept apart from other teams' machines

Condenser isn't certified for sensitive data, so real patient records would need extra safeguards first.

Q3. The data file was world-readable. Why is this not desirable? Did we change it, how and why?

Anyone on the machine could read the patient file. Now only we can.

Before	read	write
Our account	yes	yes
Our group	yes	no
Everyone else	**yes**	no

chmod

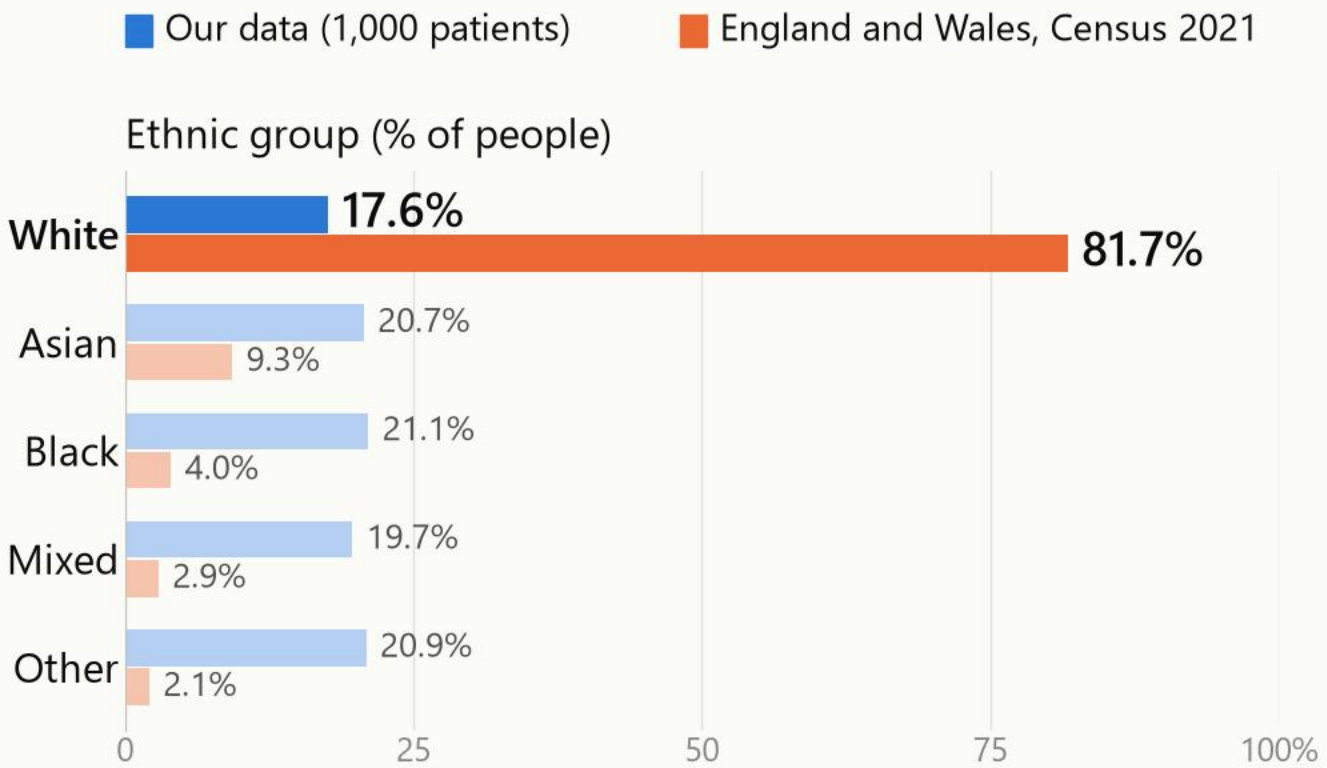
After	read	write
Our account	yes	yes
Our group	no	no
Everyone else	no	no

```
-rw-r--r--. 1 cc cc 94198 Oct 26 16:44 dig_health_hub_amr.csv
$ chmod 700 data
$ chmod 600 data/dig_health_hub_amr.csv
-rw-------. 1 cc cc 94198 Oct 26 16:44 dig_health_hub_amr.csv
# our script also sets umask 027, so the files it writes start closed to others
```

Q4. Why would you want a representative data set for machine learning?

A model is only reliable for the kinds of patients it was trained on

Ours has far fewer White patients than England and Wales, and no one under 20 or over 79





So the model knows nothing about patients outside these ages

Q5. Why would we want to use synthetic data instead of actual patient data?

Synthetic data lets us build models without putting real patients' records at risk

A real patient's record

Name	[redacted]
NHS number	[redacted]
Date of birth	[redacted]
Home postcode	[redacted]
Diagnosis	
Test results	

A real person. Protected by UK GDPR, and they may not have agreed to this use.

A synthetic record, like ours

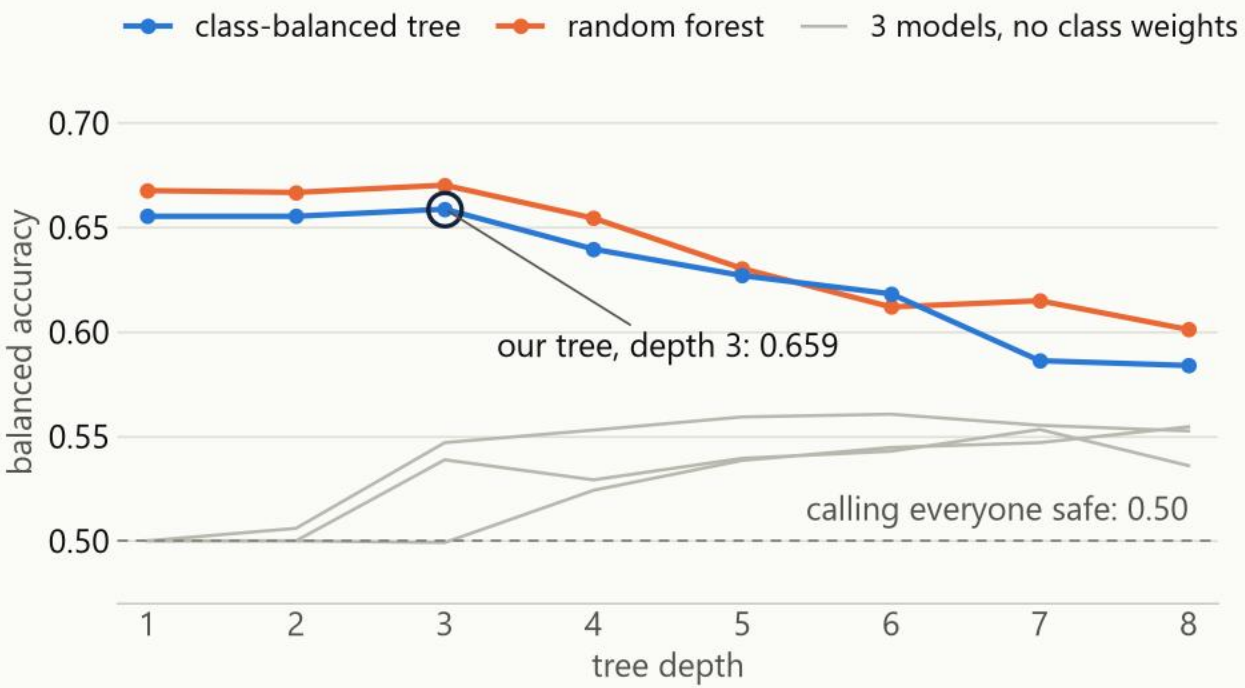
Every field is made up, even the ID and date of birth

Patient ID	
Date of birth	
Sex, ethnicity	
Deprivation	
NHS trust	
Diagnosis code	
Test results	

Nobody is behind it, so there's far less privacy risk, even on a shared cloud machine.

Q6. What is our final tree model, and why is it the most optimal?

Depth 3 was the best single tree; deeper trees did worse



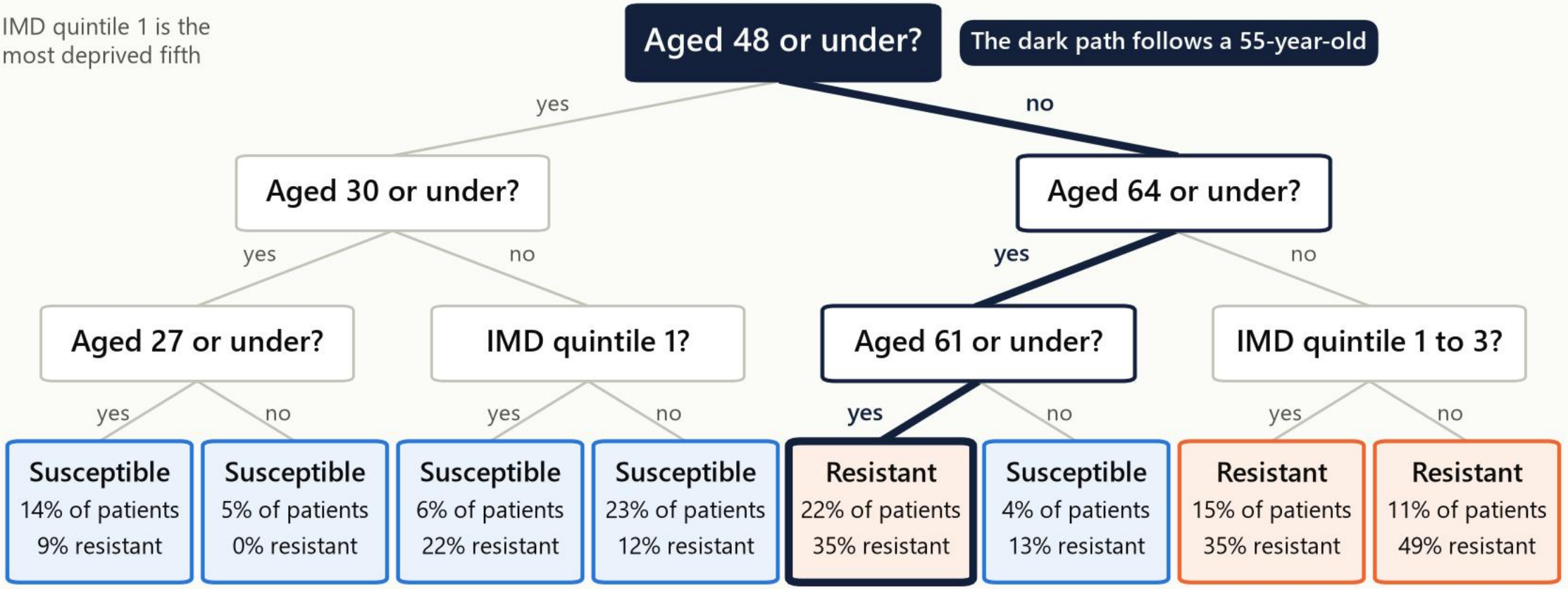
Balanced accuracy	
Calling everyone safe (76% accuracy)	0.50
Our tree, depth 3	**0.659**
Random forest, 200 trees 200 trees vote, so it can't be read like one tree	0.670

Q6. What is our final tree model, and why is it the most optimal?

Our final model is a class-balanced tree, depth 3

Patients over 48 are flagged resistant, except ages 62 to 64. Those 48 or under are susceptible.

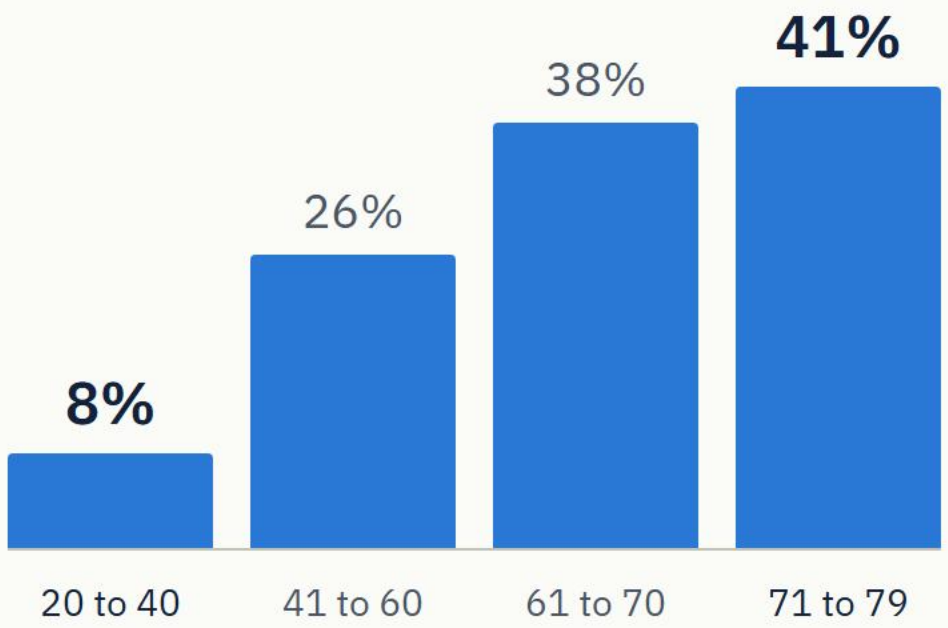
IMD quintile 1 is the most deprived fifth



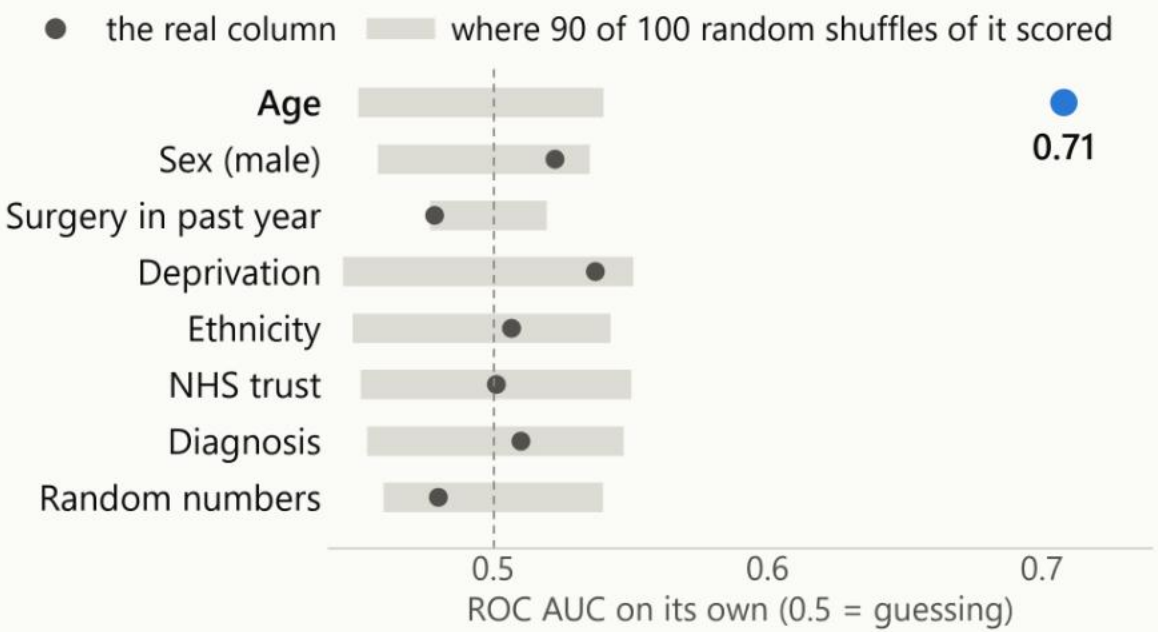
Q6. What is our final tree model, and why is it the most optimal?

Resistance rises from 8% at 20–40 to 41% at 71–79. No other factor beat random noise.

Resistant, by age



Each factor on its own



Q7. What other modelling techniques could we use, and why?

Logistic regression matched our tree, and puts a number on the age effect

1.5×

the odds of resistance for every extra 10 years of age

It fits risk that climbs smoothly with age, where a tree can only make steps. A neural network isn't needed for 1,000 patients and one clear risk factor.



Q8. How did we work as a team?

We worked in two pairs that covered for each other

Machine and script

Bobby — Cloud machine, environment and ReadMe

↕

James — Model script

Analysis and video

Ada — Explored the data and chose how to evaluate the models

↕

Eric — Led the video

A 15-minute check-in every evening

Plan, role guides and work-session log in one GitHub repo

Q8. How did we work as a team?

Three problems, and what we did about each

Problem

Login details reached only one of us, with no username

↓

What we did

Shared that login as instructed, and found the username in the organisers' access guide

Problem

No git on the cloud machine

↓

What we did

Edited on laptops, pushed to GitHub, fetched files with curl

Problem

Our first model flagged nobody as resistant

↓

What we did

Switched to balanced accuracy and class weights

Age is the only clear risk factor

Our final model is a depth-3 class-balanced decision tree,
with balanced accuracy 0.66

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`github.com/adellama75/cluster-challenge-ucl`